The Personalization Threshold: Predicting When Hyper-Personalized Advertising Becomes Intrusive

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***Abstract*—Ad platforms now tailor advertisements using demographics, interests, browsing behaviour, purchase history, location and other contextual signals. Personalization can make an ad more relevant and engaging, but using too much personal information, or the wrong kind, can raise privacy concerns and make an ad feel intrusive. For a marketer, the real question is how much personalization to apply to a given consumer.**

This study proposes a machine-learning framework called the Personalization Threshold. It estimates the highest level of personalization that can be applied to an individual before the gains from relevance and engagement are outweighed by privacy or intrusiveness risk. The framework draws on user behaviour, privacy preferences, advertisement characteristics and user–advertisement interactions, and it will be validated through a planned experiment that varies the level of personalization. Work so far covers data auditing, literature review, variable identification and construction of the analytical framework. Logistic Regression and tree-based algorithms are proposed to predict engagement and intrusiveness, and an individual-level decision rule then recommends a personalization intensity.

***Index Terms*—Personalized Advertising, Privacy Concern, Intrusiveness, Digital Marketing, Machine Learning, Consumer Behaviour, Advertisement Engagement**

I. Introduction

Digital advertising has moved from mass communication toward messages aimed at individuals. Platforms can draw on demographics, browsing activity, purchase history, interests, location and behavioural signals to make an ad more relevant to one person. The economic logic is simple: an ad that matches a consumer's needs and interests should get more attention and more engagement.

Personalization also works against itself. The same information that makes an ad relevant can show consumers how much an advertiser knows about them. An ad built on a broad interest may look helpful, but one that refers to a recent purchase, a location, a personal activity or several behavioural signals at once may feel intrusive.

Existing research treats personalized advertising as a trade-off between personal relevance and privacy-related costs. De Keyzer, Dens, and De Pelsmacker [1] found that perceived personalization raises perceived personal relevance, and that relevance in turn improves responses such as brand attitude and click intention. Other advertising research shows that highly personalized communication can create perceived intrusiveness and psychological reactance under certain conditions.

Marketers therefore face a decision problem. A uniform personalization strategy assumes every consumer tolerates the use of personal information to roughly the same degree, yet a privacy-sensitive consumer and a privacy-tolerant one may react very differently to the same ad. This study treats personalization as an individual-level prediction and optimization problem. The question moves from whether personalized advertising works to what the highest appropriate level of personalization is for a particular consumer. The goal is to estimate a personalization threshold, the point beyond which extra personalization adds too little engagement to justify the risk of perceived intrusion or an adverse consumer response.

A. Scope and Objectives

The primary objective is to build a predictive framework that recommends how much to personalize an advertisement for an individual consumer. The specific objectives are:

1) To build consumer privacy profiles from available privacy preferences, consent information and behavioural characteristics.
2) To measure or approximate the relevance between consumer behaviour and advertisement content.
3) To examine how personalization intensity, privacy sensitivity and advertisement engagement relate to one another.
4) To develop machine learning models that estimate engagement and intrusiveness risk at different personalization levels.
5) To identify an individual-specific personalization threshold, meaning the highest level of personalization that remains appropriate for that consumer.
6) To turn model predictions into an interpretable marketing decision rule that helps advertisers balance effectiveness against privacy concerns.

The framework's output is a recommendation on how aggressively an advertisement should be personalized, which goes beyond predicting whether a consumer will click it.

II. Related Work

Prior work separates actual personalization, the information technically used to customize an ad, from perceived personalization, whether the consumer recognizes the ad as personalized

[5]. The distinction matters because using more data does not guarantee that consumers will see an ad as more personalized.

De Keyzer et al. [1] studied how perceived personalization shapes consumer responses on social networking sites. In their first study (202 participants), higher perceived personalization increased perceived personal relevance, which in turn improved brand attitude and click intention. Against their initial expectation, personalization was linked to lower perceived intrusiveness. Their second study (264 participants) used several personalization elements. Perceived personalization again raised personal relevance, and its effect on perceived intrusiveness was not statistically significant. The link between personalization and intrusiveness is clearly more complicated than the assumption that more personalization always makes consumers uncomfortable.

The type and sensitivity of the information used also matter. White et al. [3] show that highly personalized communication can trigger psychological reactance when consumers feel marketers have gone too far with their personal information. Tucker [4] shows that perceived privacy control shapes how consumers respond to personalized advertising.

This leaves a gap. Most of the literature looks at average consumer responses to different personalization conditions, while consumers differ a great deal in privacy sensitivity, behavioural patterns and willingness to trade personal information for more relevant ads. A level of personalization that suits one consumer may be too much for another. This study extends the personalization–privacy literature by treating the problem as an individualized predictive decision. Instead of estimating only the average relationship between personalization and response, it predicts a consumer-specific threshold and recommends the maximum appropriate degree of personalization.

III. Methodology

The method has four stages: data integration, consumer and advertisement feature engineering, predictive modelling, and personalization-threshold estimation.

A. Data Sources and Current Data Audit

The main dataset under examination is the User Behavior & Advertisement Analytics Dataset [2]. Its documentation describes more than 350,000 records across nine interconnected tables covering user demographics, browsing behaviour, privacy preferences, advertisement characteristics and advertisement interactions. The tables most relevant to this study are:

- **UserTable:** consumer demographic attributes.
- **UserBehaviorTable:** browsing history, interaction information and purchase-related data.
- **Privacy-related features:** data-sharing permissions, privacy settings and user consent.
- **AdCharacteristicsTable:** advertisement text and targeting criteria.
- **User Interaction Table:** individual interactions between users and advertisements.
- **Interaction Metrics:** advertisement-level measures such as likes, shares, comments, clicks, impressions and conversions.

After removing metadata and embedded header rows from the CSV exports, the working data hold about 50,000 user profiles, 50,000 privacy records, 300,000 user–advertisement interaction observations and 60 advertisements.

The initial audit turned up problems that need fixing before modelling. Some user identifiers in the interaction data do not map to records in the user-profile table, and repeated identifiers appear in parts of the behavioural data. Referential-integrity validation and construction of a clean user-level analytical table are therefore part of the current preprocessing stage.

A second limitation concerns the advertisement targeting field, which mostly holds age and geographic criteria. The consumer table does not consistently contain matching attributes, so the field cannot serve as a reliable ground-truth measure of personalization intensity on its own. That is why additional personalization features and a controlled validation stage are needed.

B. Feature Engineering and Variable Construction

1. Consumer privacy profile. Privacy sensitivity will start from three available variables: data-sharing permission level, user-consent status and privacy-setting information. Each will be encoded only after its direction and category meanings have been checked. A composite Privacy Sensitivity Score may then be built, or users may be segmented into low, medium and high privacy-tolerance groups.

2. Behavioural relevance. To estimate how relevant an ad is to a user, behavioural information will be compared with advertisement content. Browsing-history text, purchase history and advertisement text will go through URL and text cleaning, tokenization, stop-word removal, lemmatization and extraction of product- or interest-related terms. A baseline TF-IDF representation with cosine similarity will then estimate how closely user behaviour matches ad content. The resulting score measures behavioural relevance and should not be read as direct evidence of perceived personalization.

3. Personalization intensity. Advertisements are conceptually divided into five increasingly individualized levels:

- **Level 0 – Generic:** no consumer-specific information.
- **Level 1 – Contextual/Demographic:** broad information such as age group or location.
- **Level 2 – Behavioural:** interests or browsing behaviour.
- **Level 3 – Historical:** previous purchase or product-history information.
- **Level 4 – Hyper-Personalized:** multiple behavioural and contextual signals used together.

For example, personalization could progress from a generic message such as “Hungry? Order dinner” to one that uses a consumer’s previous food preference and current context.

The observational dataset does not contain these five levels, so they cannot be treated as existing labels. A controlled survey

or advertisement experiment is proposed to measure consumer reactions to progressively personalized ads directly.

4. Outcome variables. Two categories of outcome are proposed. Behavioural outcomes from the existing dataset are click, like, share, comment, conversion and other available engagement measures. Perceptual outcomes from the planned validation experiment are perceived relevance, perceived personalization, perceived intrusiveness, and click or purchase intention.

A direct intrusiveness measure is needed because engagement alone cannot be taken to represent privacy comfort. A consumer may interact with an ad and still find its use of data intrusive.

C. Proposed Hypotheses

Based on the literature and the project framework, four initial hypotheses are proposed.

- **H1:** Increasing advertisement personalization will generally increase perceived personal relevance.
- **H2:** Consumer privacy sensitivity will moderate the relationship between personalization intensity and perceived intrusiveness.
- **H3:** The relationship between personalization and consumer response will vary across consumers, and the optimal personalization level will depend on individual privacy and behavioural characteristics.
- **H4:** Beyond an individual-specific personalization threshold, additional personalization will provide too little engagement benefit relative to its predicted privacy or intrusiveness risk.

H2 does not assume that personalization must always increase intrusiveness. Existing empirical findings are mixed, so how consumer characteristics interact with personalization intensity is itself an open research question.

D. Models and Analytical Workflow

Stage 1: Engagement prediction. The first task estimates the probability that a user engages with an advertisement. Likely predictors are consumer privacy variables, the behavioural relevance score, advertisement characteristics, user characteristics and, when available, personalization level. A Logistic Regression model will serve as an interpretable baseline. It will then be compared with nonlinear models such as Random Forest and Gradient Boosting / XGBoost, which can pick up nonlinear relationships and interaction effects among privacy sensitivity, advertisement relevance and personalization.

Stage 2: Intrusiveness prediction. Once perceptual data are collected, a separate model will estimate perceived intrusiveness. Because it will likely be measured on an ordered Likert scale, both continuous prediction and ordinal classification will be evaluated, rather than converting the measure into an arbitrary binary variable.

Stage 3: Personalization threshold estimation. For each consumer i and personalization level p, the system will estimate the expected engagement probability and the expected intrusiveness risk. A simplified marketing utility function is:

$$\begin{aligned}\mathrm{Utility}(i,p) = \mathrm{ExpectedEngagement}(i,p) \\ - \lambda\,\mathrm{IntrusivenessRisk}(i,p)\end{aligned} \tag{1}$$

where λ is the relative business cost assigned to consumer discomfort or privacy risk.

A consumer's recommended threshold is the highest personalization level that improves expected marketing value without pushing predicted intrusiveness risk past an acceptable limit. The final system would then give an actionable recommendation such as "Consumer A – Recommended Level 2: Behavioural Personalization" in place of "Consumer A – 72% probability of clicking." This difference is the main business contribution of the project.

IV. Evaluation Metrics

Predictive performance and business usefulness will be evaluated separately.

For binary engagement prediction, the proposed metrics are accuracy, precision, recall, F1-score and ROC-AUC. The threshold decision needs probability estimates, so probability calibration will also be examined. For continuous intrusiveness or relevance prediction, the metrics are Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2. To limit data leakage, observations from the same consumer should not be split indiscriminately across training and testing sets, so user-level grouped validation is preferred when a consumer has multiple observations.

From a business perspective, the recommendation system will be compared with two simple benchmarks:

1) Generic advertising for every consumer.
2) Maximum personalization for every consumer.

The model will be considered valuable if it improves predicted engagement while keeping intrusiveness risk lower than a blanket maximum-personalization strategy.

V. Current Progress and Limitations

The project is in the research-design and data-preparation stage. Work completed so far:

- Definition of the personalization-threshold business problem.
- Identification of the personalization–privacy trade-off in existing literature.
- Selection and initial examination of the advertisement analytics dataset.
- Identification of the tables needed for user, privacy, advertisement and interaction analysis.
- Initial data-quality and referential-integrity assessment.
- Development of the proposed personalization-level framework.
- Identification of candidate features, target variables, models and evaluation metrics.

The main limitation is that the advertisement-interaction dataset has no direct perceived-intrusiveness label. Privacy preferences and engagement behaviour cannot establish on their own whether an ad was perceived as intrusive, so the project

should not claim that the existing dataset predicts creepiness or intrusiveness. A direct measurement stage using controlled advertisement variants and consumer ratings is proposed to establish ground truth. The behavioural dataset would then supply scale and interaction information, while the controlled study would supply the perceptual variables needed to validate the Personalization Threshold concept.

VI. Implementation Tools

The project will be implemented mainly in Python. Pandas and NumPy will handle data manipulation and integration, and NLTK, spaCy and scikit-learn's TF-IDF tools will handle text preprocessing and feature extraction. Models will be built with scikit-learn, with boosting techniques evaluated where they fit, and feature-importance and SHAP-based analysis can support interpretation. Matplotlib and related Python libraries will visualize distributions, model performance, privacy segments, engagement patterns and personalization thresholds. Development will run in Jupyter Notebook or Google Colab so the whole analytical pipeline is documented and reproducible.

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